

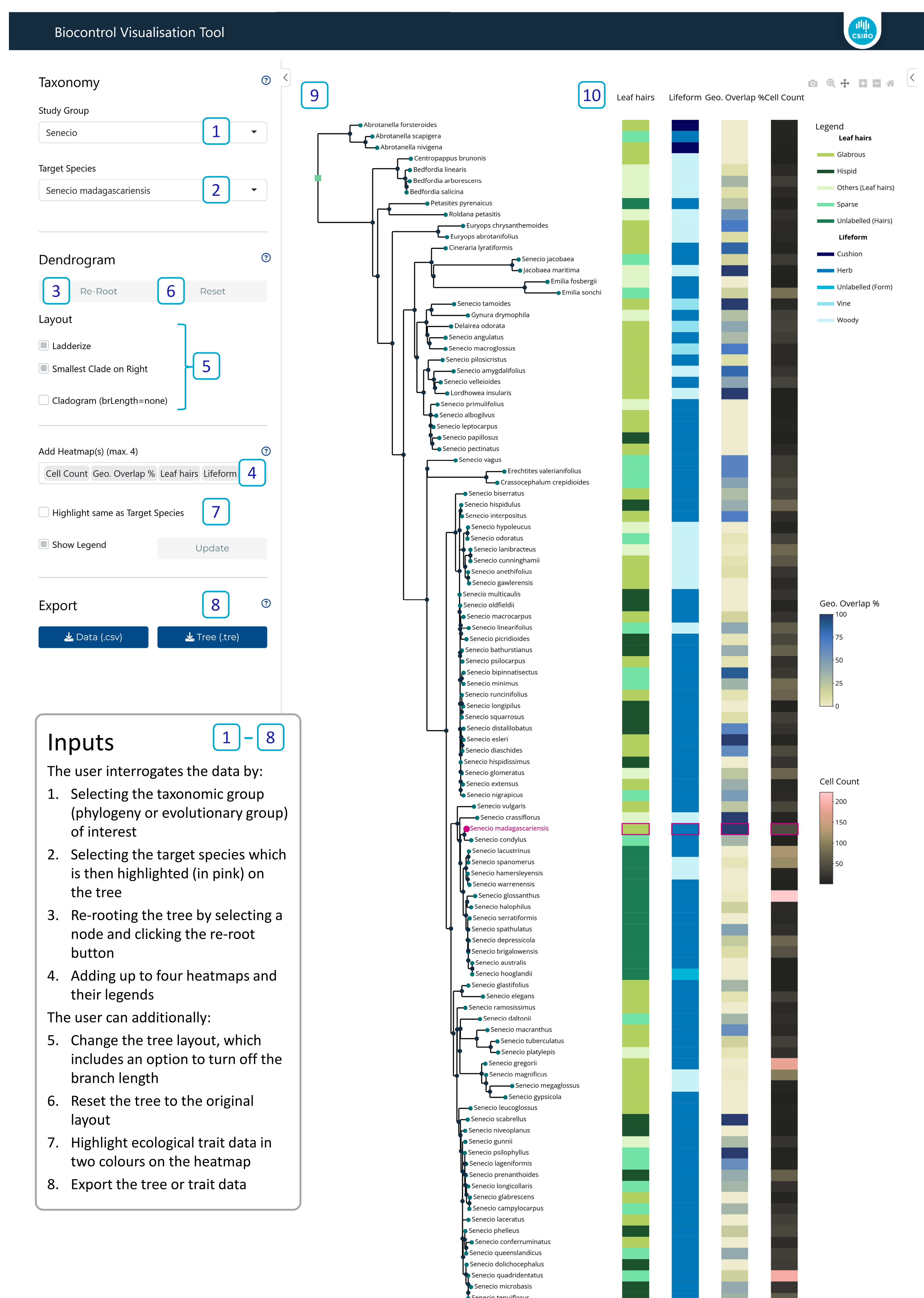
A Prototype Visualisation Tool for Biocontrol Researchers

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Biological control is the practice of releasing herbivores or pathogens to manage invasive species. Investigating whether potential biocontrol agents will suitably control the invasive (target) species without affecting important native or horticultural (non-target) species requires carefully designed host specificity tests. Testing a candidate control agent against every possible host species is time consuming and costly, so it is important for researchers to compile a manageable test list of the species that are most at-risk. Of particular interest are species that are co-located, closely related with, and ecologically similar to the target, because e.g., herbivores find closely related plants similarly palatable. To construct their test lists, researchers frequently have to conduct lengthy literature reviews, consultations with taxonomists, and phylogenetic and spatial analyses.

Existing tools for the visualisation of relevant data in an evolutionary context do not provide the functionality required for risk analysis and are not designed as decision support tools for biocontrol researchers. We have developed a prototype visualisation tool to address this gap. We use Shiny, a flexible web framework package written in R, to visualise relevant data as an ensemble and dynamically explore it from the perspective of the target species. The tool aims to enable the data to be queried in a way that informs the development of a test list of at-risk non-target species that could potentially be affected by the biocontrol agents.



Phylogenetic Tree

The dendrogram presents the evolutionary relatedness of each species. We use the *ape* package for handling and manipulating the tree data and a combination of *ggtree*, *ggplot2* and *plotly* to generate the dendrogram.

Re-Rooting

When the uploaded tree is unrooted or arbitrarily rooted, it is important to have the ability to re-root the tree. We have implemented outgroup rooting that allows the user to pick an internal node upon which to re-root. When changing the tree root, or the target species, the heatmaps update responsively to present the appropriate target versus non-target values. This enables the researchers to identify the species that are closely related and ecologically similar to the target.

Heatmaps

A range of phylogenetic, ecological, and geographical trait metrics are displayed in the heatmaps. They are built in *ggplot2*, and *plotly* is applied to provide interactivity and align them with the dendrogram as subplots.

Colour

Colour allows the user to summarise and rank information visually and quickly for each species with respect to the target.

We choose colours that are distinguishable, colour blind friendly and perceptually coherent. We use the *scico* package for the continuous scales, which meets the above criteria. For nominal data with more than five classes, we collate the low frequency categories into an “others” group. This ensures distinct colours can be assigned.

The tool has an option to highlight ecological trait data in two colours: one for the traits with the same value as the target species and grey for all others. The two perceptually contrasting colours are chosen to enable species of interest to be quickly identified.

Future Work

As a prototype, development is ongoing and will be guided by user feedback and suggestions for additional functionality. Future capability includes:

- The ability for researchers to upload datasets.
- The highlighting of selected test list species, to indicate how they are distributed across the phylogeny.
- The visualisation of other rationales for selection such as taxonomic information (same genus as target species, same subfamily, etc.).
- Geographic visualisations showing species distributions.
- Incorporation into an analysis pipeline.

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